

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester VI Examination, 2022-23

Zoology

Paper: ZOB-605 — Immunology and Parasitology

Time: Three hours

Full Marks: 70

Note: Answer three questions from Section-A (including Question No. 1 which is compulsory) and three questions from Section-B (including Question No. 1 which is compulsory). Use a separate answer book for each section. The figures in the right-hand margin indicate marks.

SECTION – A
(Immunology) (Marks: 35)

Question 1. Answer the following parts:

(a) State whether the following statements are True or False. Justify your answer in 2–3 sentences each: [4 × 2 = 8]

- (i) Type II hypersensitivity is mediated by immune complex.
- (ii) B-cell epitopes are always presented with MHC.
- (iii) Herd immunity is achieved in highly contagious disease when 40% of population is vaccinated.
- (iv) Graves' disease affects the thyroid gland.

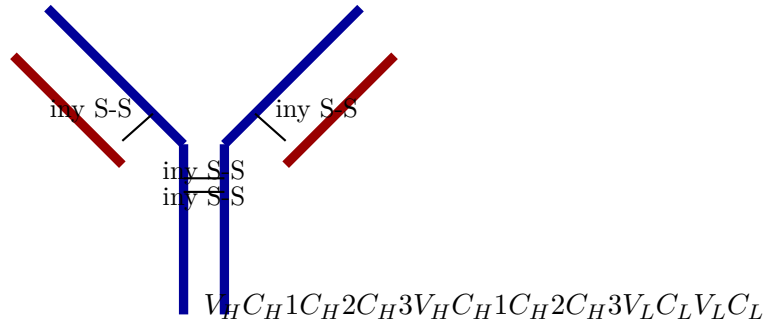
(b) Define the following: [3 × 1 = 3]

- (i) M cells
- (ii) Immunodeficiency
- (iii) Adaptive immunity.

Question 2. Describe the mechanism of antigen processing and presentation via MHC class I and class II to T lymphocytes. [12]

Question 3. Answer the following:

- (a) Discuss the properties of an ideal vaccine. [5]
- (b) Discuss the structure of immunoglobulin and describe the function of IgG, IgA and IgM. [7]



Question 4. Write short notes on the following:

[3 × 4 = 12]

- (a) Lymphocytes
- (b) Hapten
- (c) Spleen.

SECTION – B (Parasitology) (Marks: 35)

Question 1. Answer the following parts:

- (a) State whether the following statements are True or False. Justify your answer in 2–3 sentences each: [2 × 2 = 4]
- (i) Individual suffering from trichinosis develops myositis, fever, periorbital oedema within a week.
 - (ii) Unembryonated eggs of *Paragonimus westermani* are coughed up and voided in sputum.
- (b) Differentiate between the following: [2 × 2 = 4]
- (i) Endotoxin and Exotoxin
 - (ii) Tachyzoite and Bradyzoite.
- (c) Define the following: [3 × 1 = 3]
- (i) Hyperparasites
 - (ii) Paratenic host
 - (iii) Zooanthroponosis.

Question 2. Discuss the asexual cycle of *Plasmodium vivax*. Comment on the clinical manifestations and control measures of the disease. [12]

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(Injected by Mosquito);
ode[draw, rounded corners, fill=orange!10, align=center] (liver) at (3.8, 2.5) Invasion of
Liver Cell
(Exo-erythrocytic cycle);
ode[draw, rounded corners, fill=orange!15, align=center] (schiz1) at (4.8, 0) Liver Schizont
& Merozoites;
ode[draw, rounded corners, fill=red!10, align=center] (rbc) at (2.5, -2.5) Invasion of RBC
(Erythrocytic cycle);
ode[draw, rounded corners, fill=red!15, align=center] (ring) at (0, -4) Ring Stage
(Trophozoite);
ode[draw, rounded corners, fill=red!20, align=center] (schiz2) at (-2.5, -2.5) Erythrocytic
Schizont
(Merozoite release);
ode[draw, rounded corners, fill=purple!10, align=center] (gam) at (-4.8, 0) Gametocytes
(Infective to Mosquito);

[->, ultra thick, blue!60!black] (sporo) to[out=0, in=120] (liver); [->, ultra thick,
blue!60!black] (liver) to[out=-30, in=90] (schiz1); [->, ultra thick, blue!60!black] (schiz1)
to[out=210, in=30] (rbc); [->, ultra thick, red!60!black] (rbc) to[out=240, in=0] (ring); [->,
ultra thick, red!60!black] (ring) to[out=180, in=-60] (schiz2); [->, ultra thick, red!60!black]
(schiz2) to[out=30, in=150] (rbc); [->, ultra thick, purple!60!black] (schiz2) to[out=120,
in=-90] (gam); [->, ultra thick, green!60!black] (gam) to[out=90, in=180] (sporo);
ode[above, font=] at (0, 4.8) Asexual Life Cycle of Plasmodium vivax;

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